

Meeting 2:OFDM+Next week plan

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Thanks to Kailong Wang for helping me understand
OFDM knowledge and answers all of my questions!

Outline

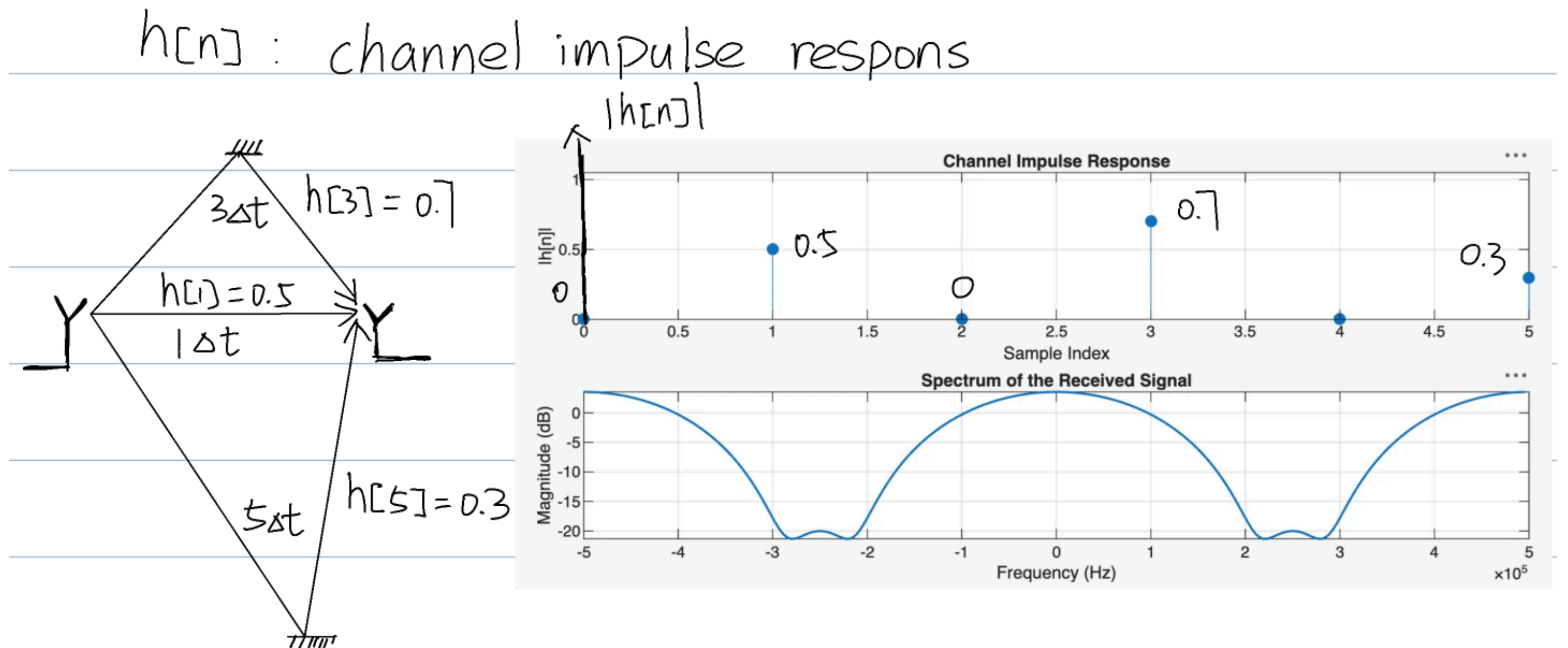
1. What obstacles prevent reliable communication in Wireless communication (with OFDM how to overcome these separately, why OFDM is good)
2. MATLAB code shows whole system about the OFDM simulation
 - Constellation of both Modulation&Demodulation(QPSK only)
 - Show where create the OFDM in the system
 - How to equalization in a real world(Cannot know H_{freq} , use $H_{\text{estimation}}$)
 - Show system impulse response(time&freq domain), $Y(\text{freq})$, Compare BER vs SNR
3. Next week plan: Papers for modeling the IRS mobile scenario

What obstacles prevent reliable communication #:will detailed explain

- Intersymbol Interference (ISI) <— —- due to Multipath delay spread#
- Frequency Selective fading <— —- due to Multipath Propagation#
- Additive White Gaussian Noise <— —- Naturally present in communication system (e.g. thermal noise of electronic device)

An easy demo to explain Multipath Propagation.

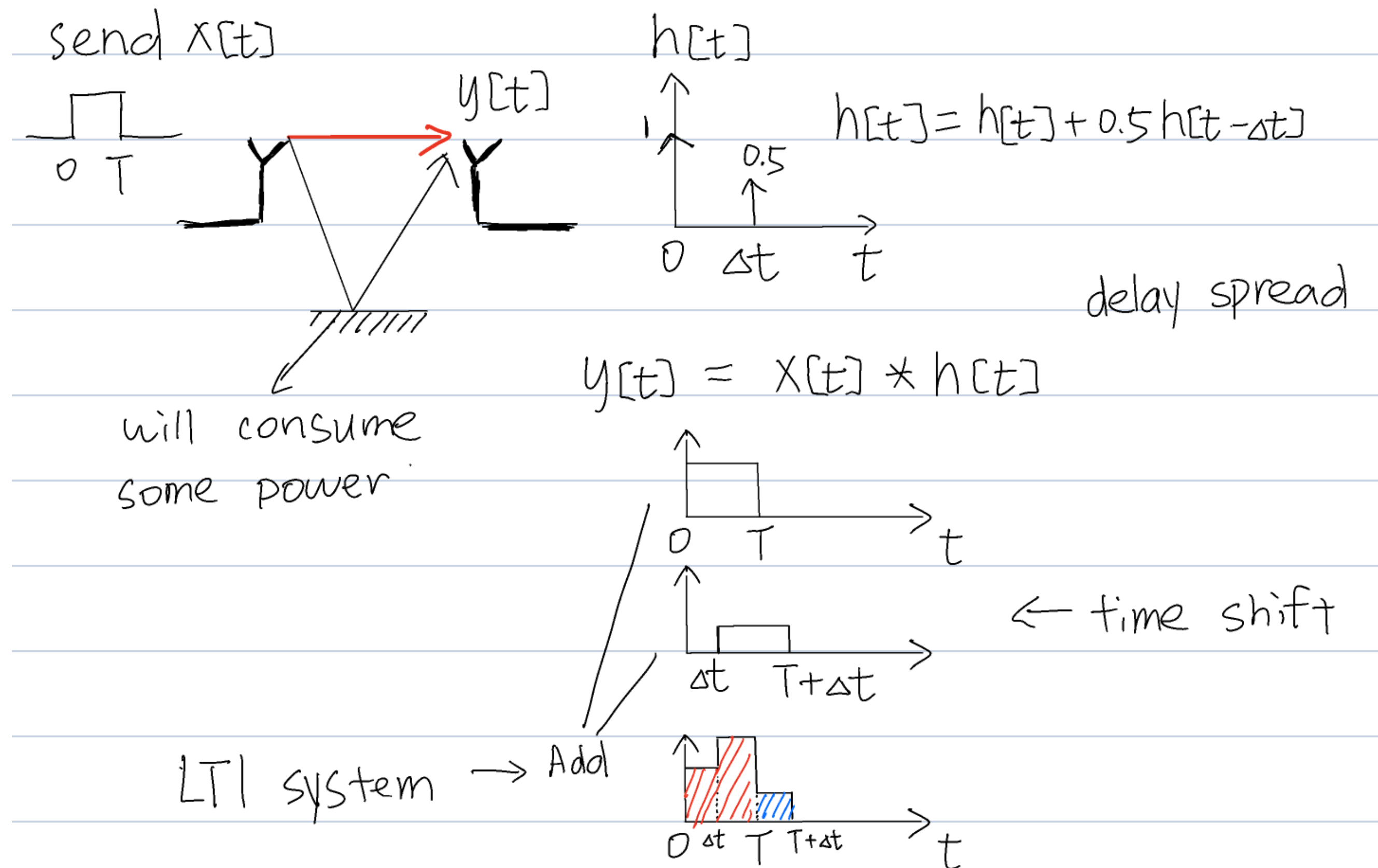
*Here just ignored phase shift. It exists in the channel.



$$h[n] = 0.5 h[n-1] + 0.7 h[n-3] + 0.3 h[n-5]$$

What obstacles prevent reliable communication

- Intersymbol Interference (ISI) <— —- due to Multipath delay spread (DS)



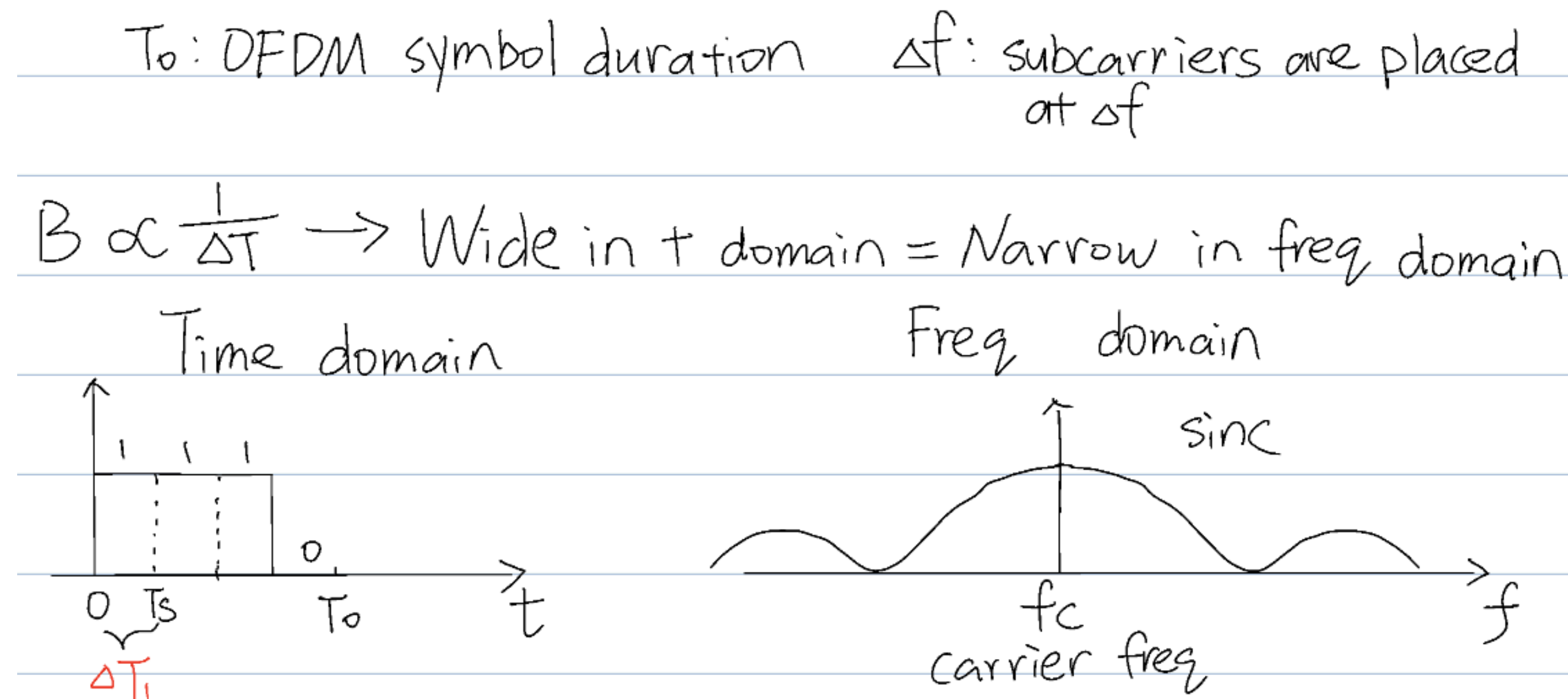
: We received : will affect next TX & RX
 → if $x[n-1]$ send before $x[n]$,
 $y[n]$ will be affected by $y[n-1]$

- Notice that ISI will affect the front signal
 — — —> If we can give a room for them and delete later, then we can fix it.
 — — —> We designed the: Cyclic Prefex !
- And if delay spread $> 0.1T_s$, ISI will happen.
 — — —> If we can make T bigger. Then it acquire bigger DS to enable ISI happen
 — — —> OFDM is good: narrow in freq
 —> wide in t domain -> big T (explain next)

$\star \rightarrow$: assume will transmit immediatly, so we should receive from $0-T$

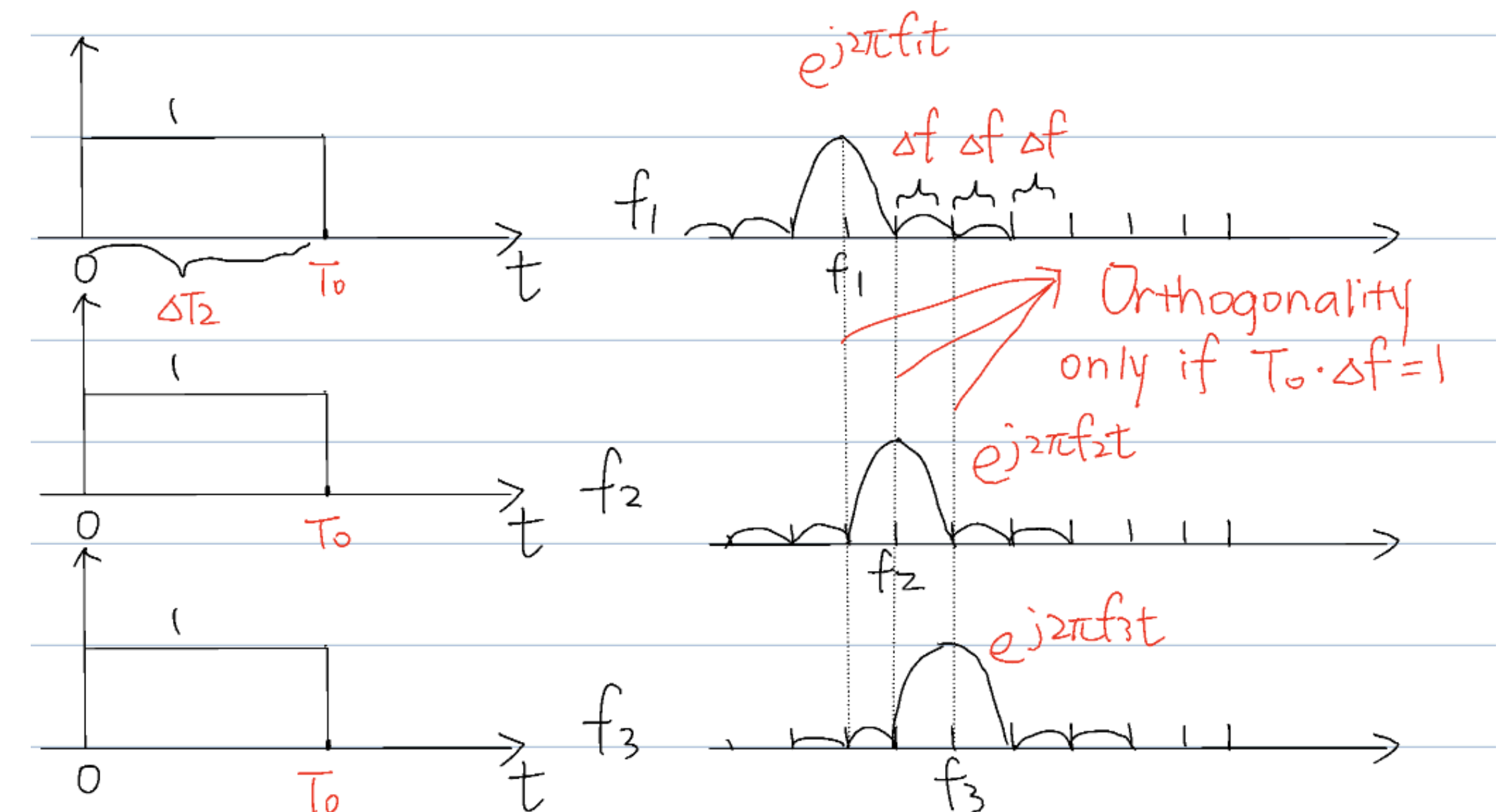
What obstacles prevent reliable communication

- Intersymbol Interference (ISI) $\leftarrow$ due to Multipath delay spread (DS)



Not OFDM

How to prove $T_0 \cdot \Delta f = 1$? Use $\langle e^{j2\pi f_1 t}, e^{j2\pi f_2 t} \rangle = 0$
 subcarriers are orthogonal
 $\Rightarrow \langle a, b \rangle = 0$ cross-correlation between any two of them is 0



OFDM

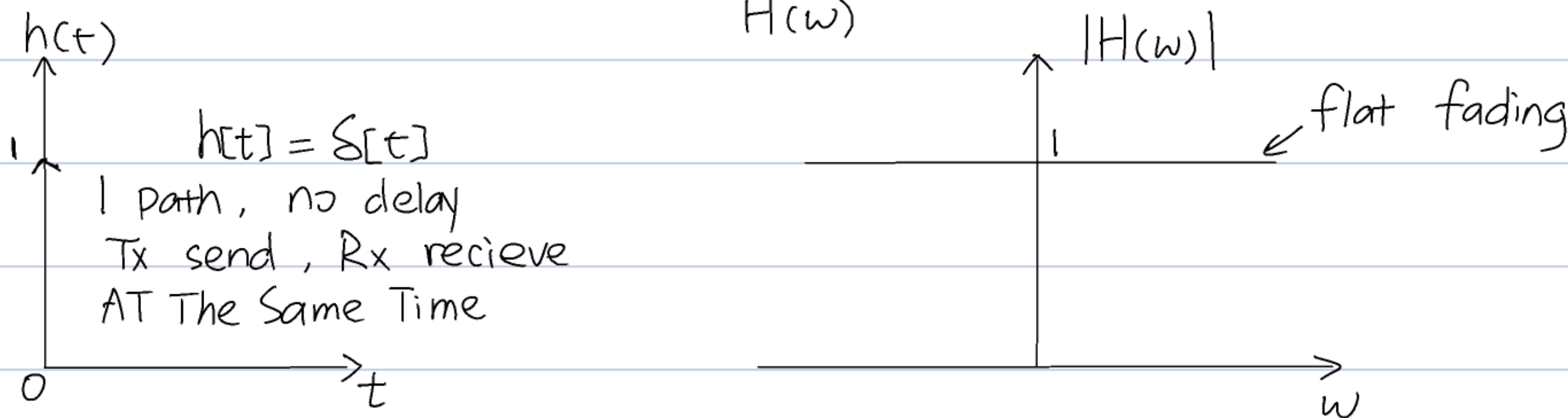
What obstacles prevent reliable communication

- Frequency Selective fading <— —- due to Multipath Propagation
- Fading: signal attenuation (e.g. multi path ; rain; obstacles)
- 2 type of fading based in freq domain(also have slow, fast fading, they are in time domain, and sth like block fading. No discuss here)
 - Flat fading: A flat line in freq domain plot, which means channel affects all freq components equally.
 - Freq Selective fading: A line in interesting shape in freq domain. Affect all part of Freq in it own way.
 - (diagram to explain next!)

What obstacles prevent reliable communication

- Frequency Selective fading <— —- due to Multipath Propagation

Flat fading : $s[t] \xrightarrow{\text{F.T.}} 1$



bandwidth

$$C = B \log_2(1 + \text{SNR})$$

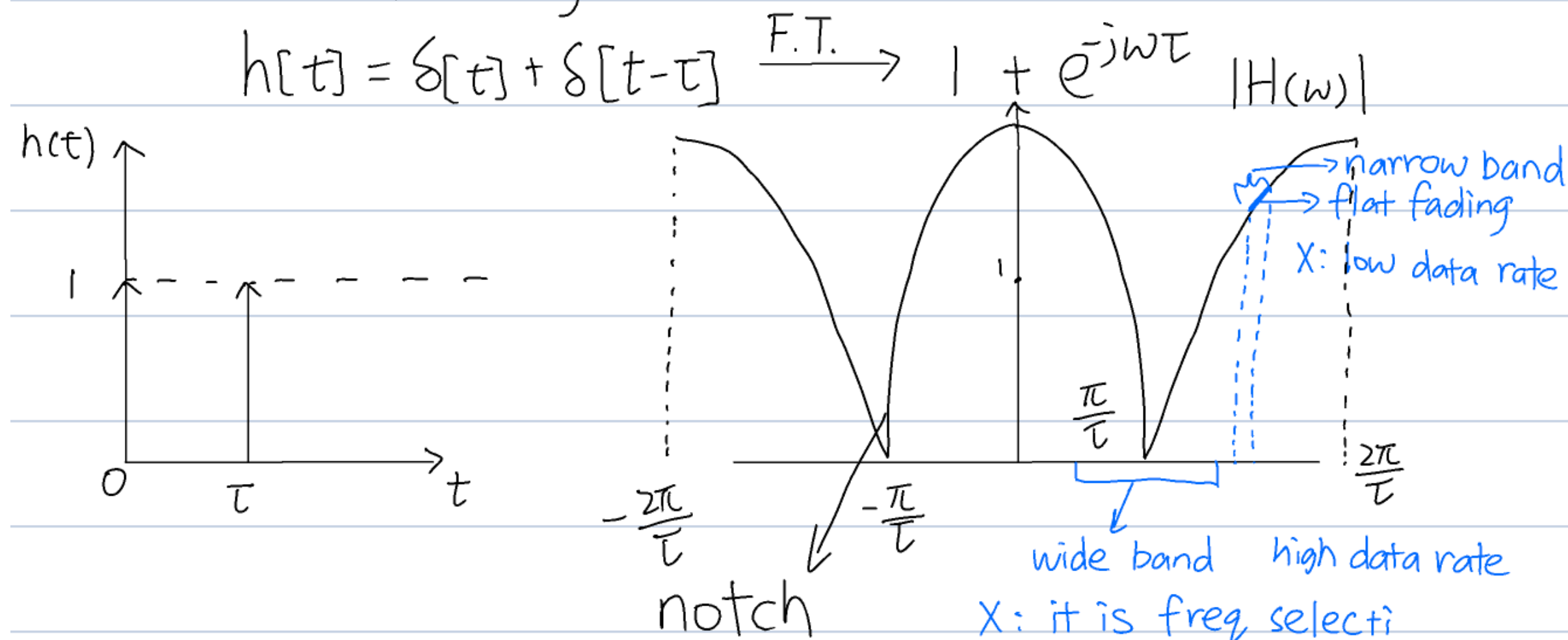
channel capacity (bits/s)

small B : low C
big B : high C

narrow band is good for free-selective fading but low data rate

wide band is good since high data rate but easily to be affected by notch.

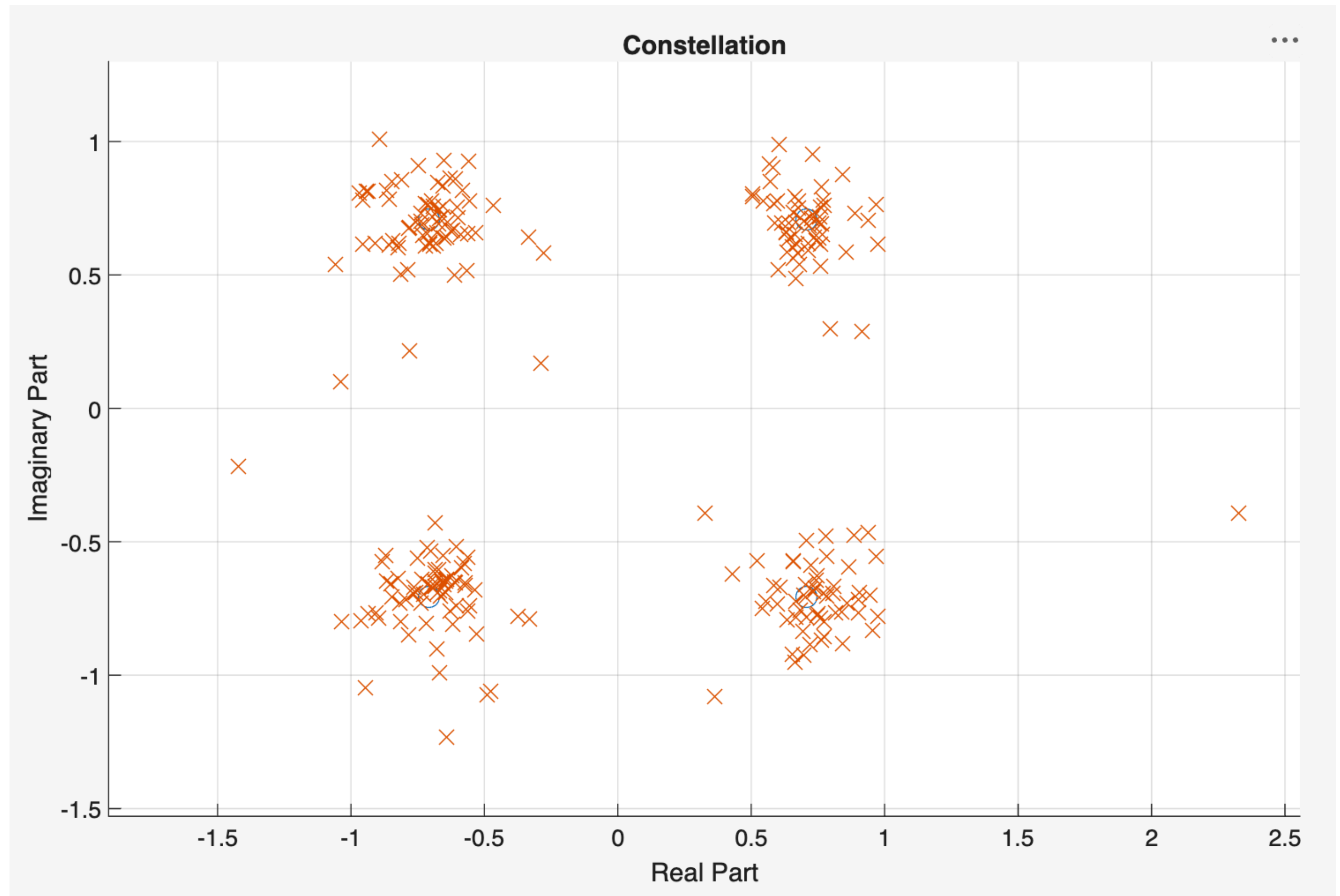
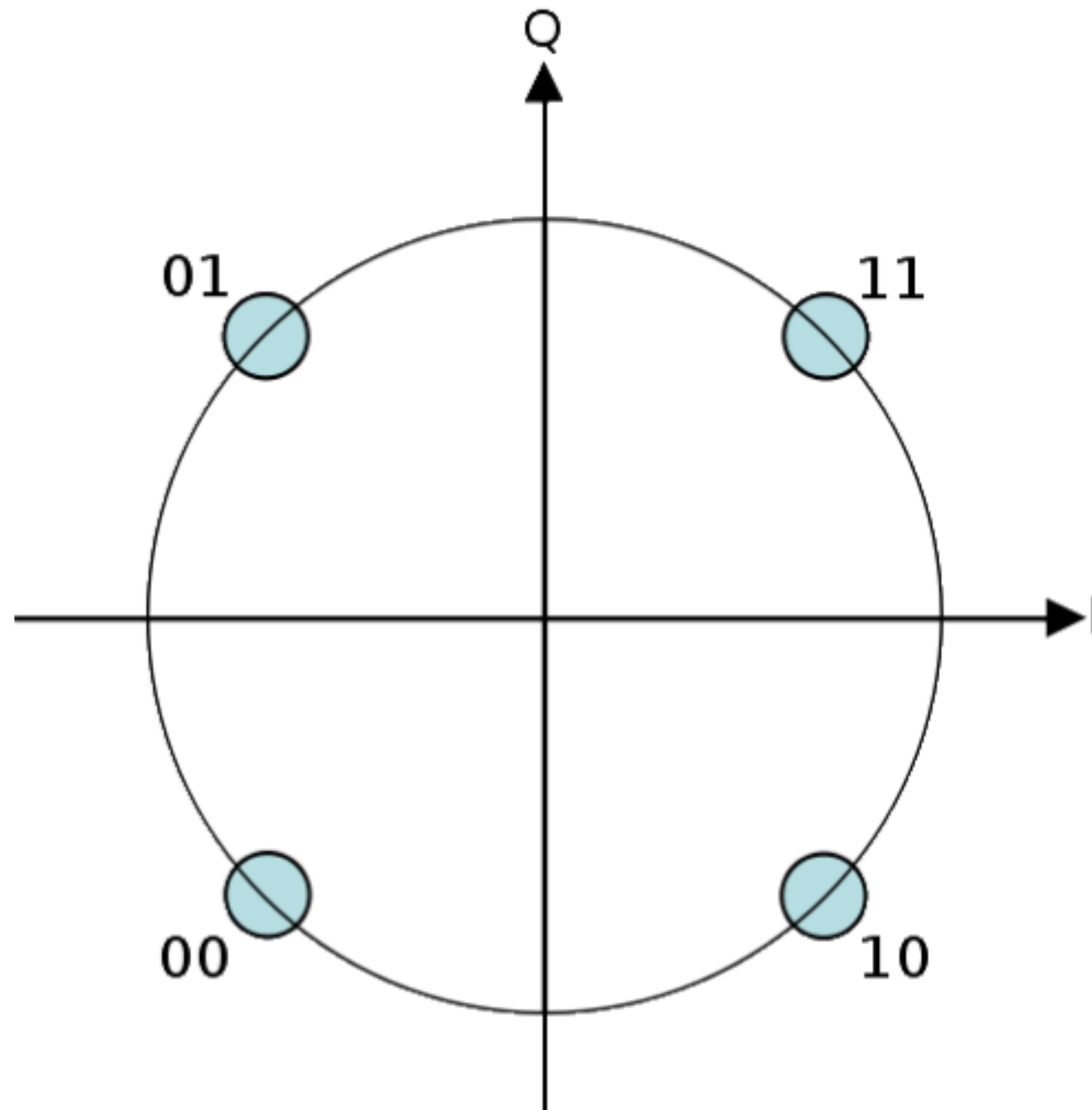
Selective fading:



What if put lots of narrow band together each narrow band can overcome notch and included high data rate
-> Then we have OFDM!

MATLAB:

- Constellation of both Mo&Demodulation(QPSK only)



MATLAB:

- Show where create the OFDM in the system

```
% Modulation  
%-----  
function [X_freq, x_cp_time] = Modulation(x, M, cpLength)  
    X_freq = pskmod(x, M, pi/4, InputType="bit");  
    x_time = ifft(X_freq);  
    cp_time = x_time (end - cpLength + 1: end);  
    x_cp_time = [cp_time; x_time];  
end
```

Where is OFDM in the code

After pskmod $\rightarrow$ We get $X[0], X[1], \dots, X[N-1]$

$x_time = \text{ifft}(X_freq)$ **creat OFDM**

$$\text{IDFT: } x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi \frac{kn}{N}}$$

$$x[n] = \frac{1}{N} [X[0] e^{j2\pi 0n/N} + X[1] e^{j2\pi 1n/N} + \dots]$$

$\uparrow$ subcarrier freq $\uparrow$ Total N subcarrier

$\Rightarrow$ OFDM in time domain $x[n]$

f_c : carrier freq

$x[n] \cdot e^{j2\pi f_c t} \rightarrow$ Then send over Air

$\uparrow$
multiple in time domain, convolution in freq domain
(freq, shift)

* f_c is not included in Code

MATLAB: In real world. To do the equalization, we need to insert x_{pilot} in x (the signal we want to send) due to the quickly changing channel. Here we just show a easy demo.

How to equalization in a real world (Cannot know H_{freq} , use $H_{\text{estimation}}$)

```
x = randi([0 1], num_of_bit_stream, 1); %random data input
x_pilot = ones(num_of_bit_stream, 1); %known input For get H_estimated_freq

y_time_noisy = Transmit_signal(x_cp_time, h_time, SNR_dB);
y_pilot_time_noisy = Transmit_signal(x_cp_pilot_time, h_time, SNR_dB);
```

```
function y_time_noisy = Transmit_signal (x_cp_time, h_time, SNR_dB)
y_time = conv(h_time, x_cp_time);
y_time_noisy = awgn(y_time, SNR_dB, "measured");
end
```

```
function [y, Y_equalization, H_estimated_freq, Y_freq] = Demodulation
y_pilot_nocp_time = y_pilot_time_noisy(cpLength+1: cpLength+N);
y_nocp_time = y_time_noisy(cpLength+1 : cpLength+N); %minus cp
```

```
Y_pilot_freq = fft(y_pilot_nocp_time);
Y_freq = fft(y_nocp_time);
```

```
%Equalization
```

```
H_estimated_freq = Y_pilot_freq ./ X_pilot_freq;
```

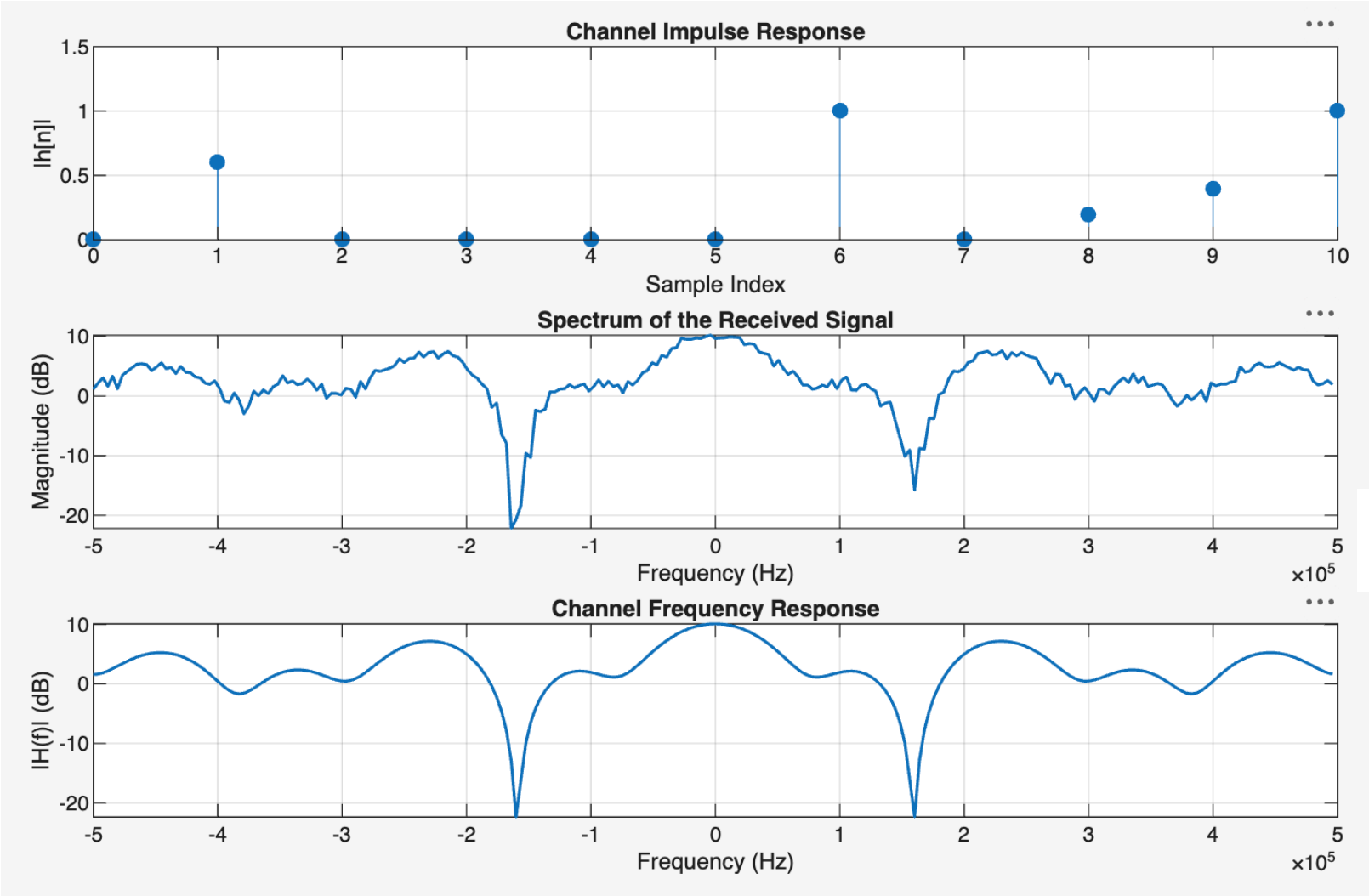
```
Y_equalization = Y_freq ./ H_estimated_freq;
```

```
y = pskdemod(Y_equalization, M, pi/4, OutputType = "bit");
```

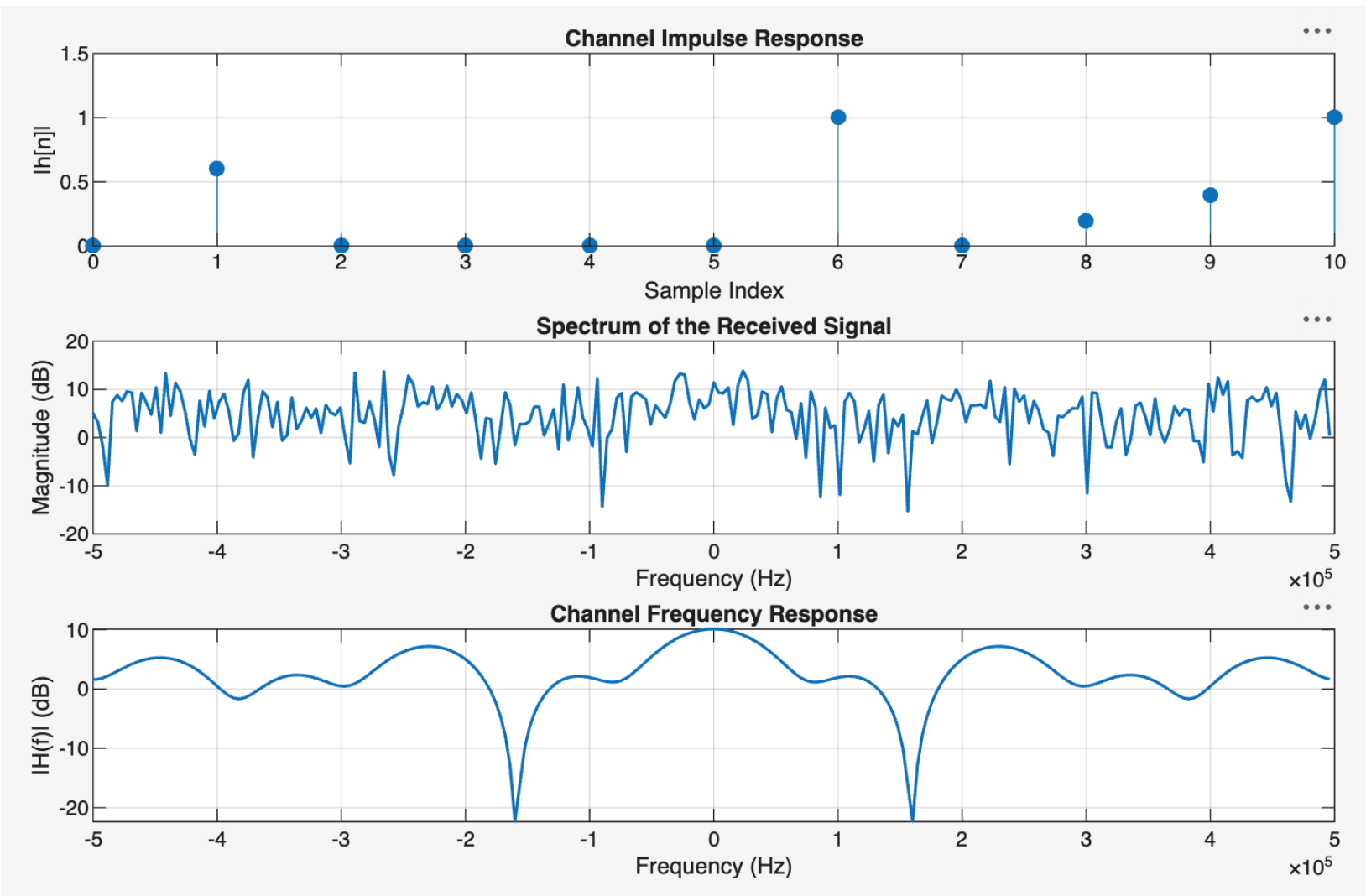
```
end
```

MATLAB:

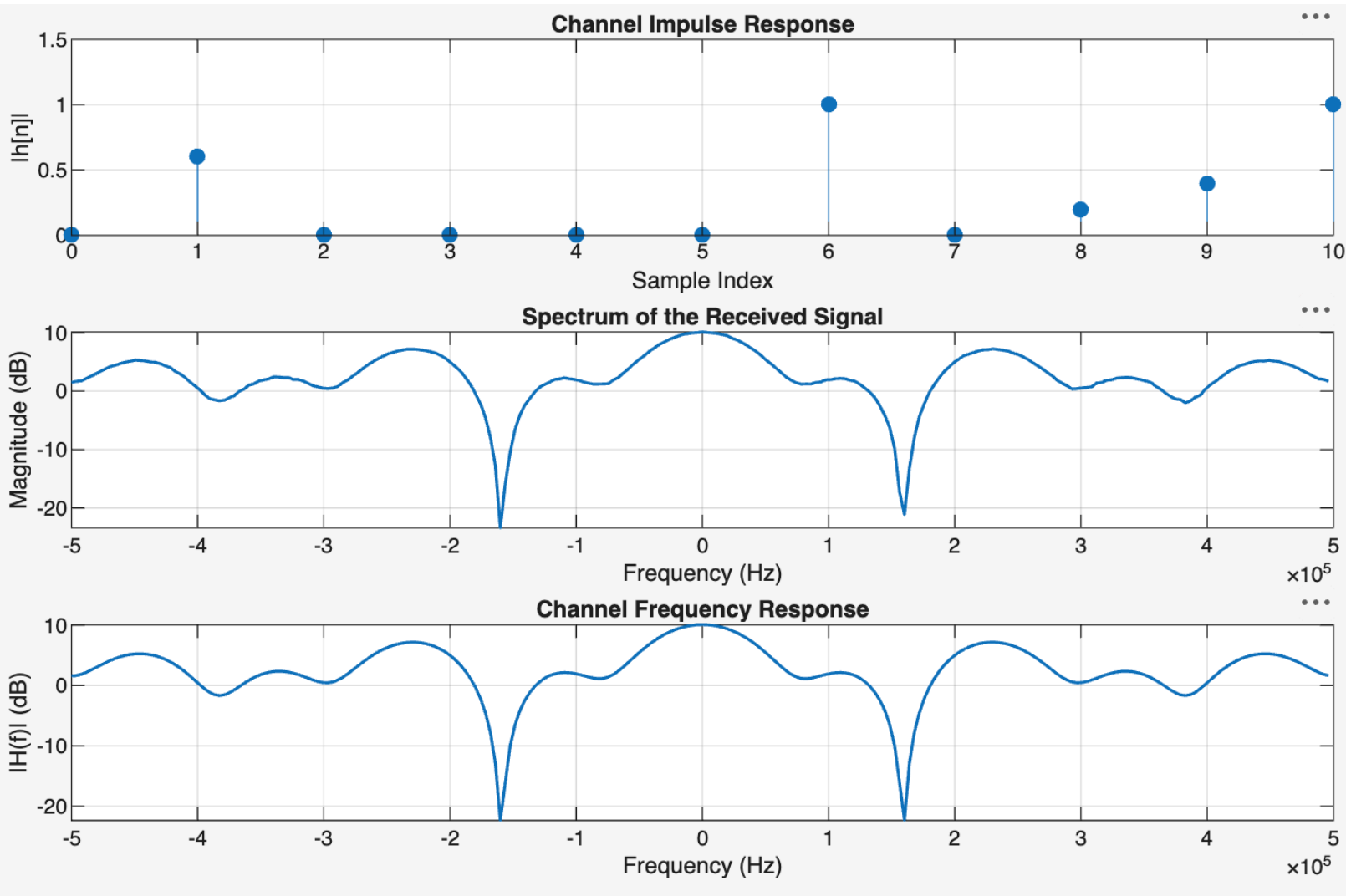
- Show system impulse response(time&freq domain), Y(freq), Compare BER vs SNR



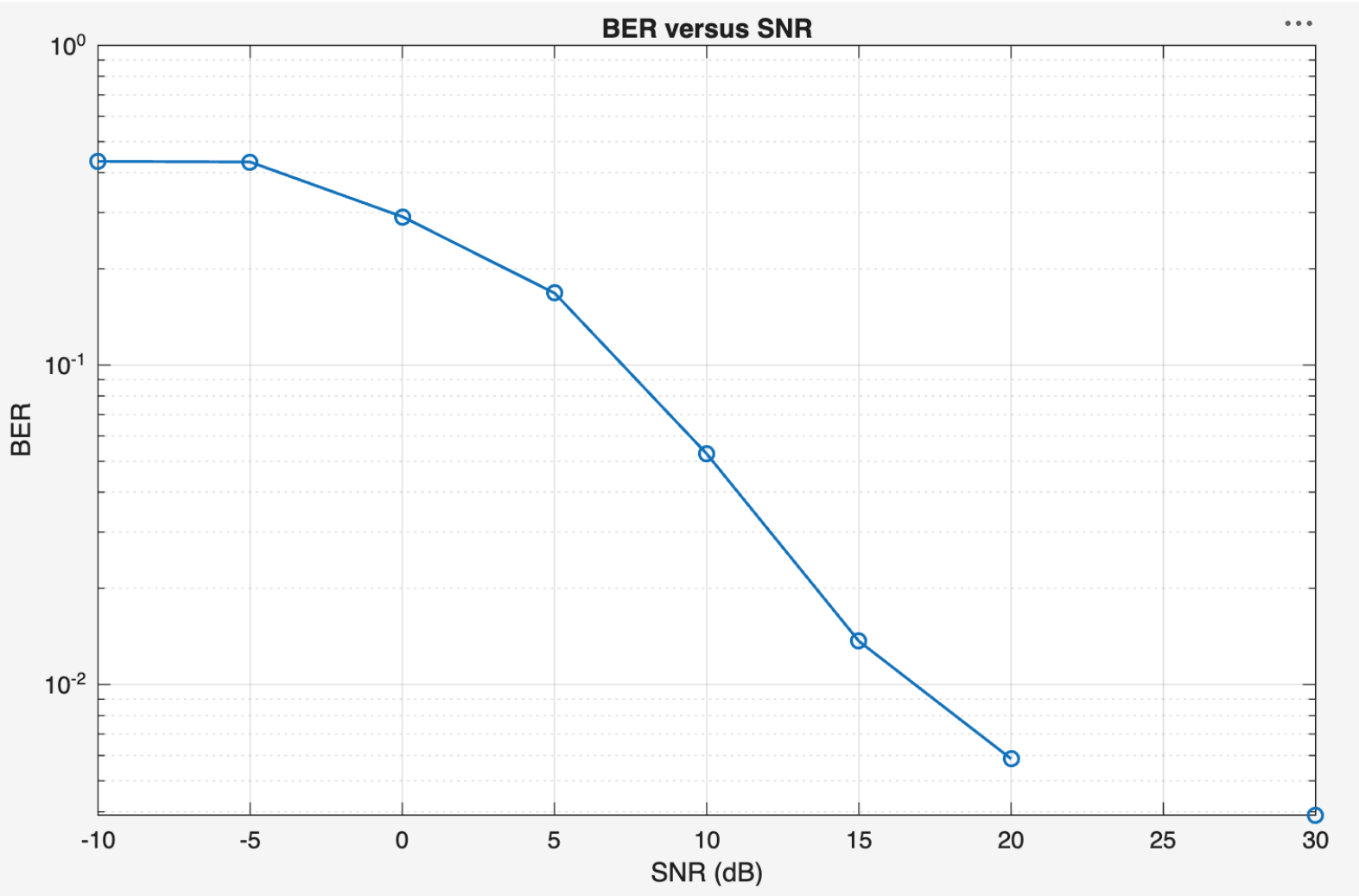
SNR_dB = 20;



SNR_dB = 0;



SNR_dB = 40;



Next week plan: Papers for modeling the IRS mobile scenario

Secure Intelligent Reflecting Surface Aided Integrated Sensing and Communication

Found this one in reference

Meng Hua, Qingqing Wu, *Senior Member, IEEE*, Wen Chen, *Senior Member, IEEE*, Octavia A. Dobre, *Fellow, IEEE*, and A. Lee Swindlehurst, *Fellow, IEEE*

Learning-Aided Adaptive Beampatterns for Secure Transmission and Enhanced Sensing in DFRC

This one for extra

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